Tensor networks and quantum inference for neuro-symbolic AI

Alex Goessmann
joint work with Janina Schütte, Maximilian Fröhlich and Martin Eigel

Weierstraß Institute for Applied Analysis and Stochastics (WIAS)

SIAM PP Conference 2026 - MS79
March 6, 2026

Outline

- Tensor networks in neuro-symbolic AI
- Quantum Inference for Neuro-Symbolic AI

Outline

- Tensor networks in neuro-symbolic AI

- Quantum Inference for Neuro-Symbolic AI

Motivation: Accounting Rule Systems

Example (Toy Accounting Rule System)

Assume you want to book one of two accounts $\mathrm{A1}$ and $\mathrm{A2}$ depending on a feature F of an invoice, based on two rules:

- Exactly one account must be booked.
- If feature F is present on the invoice, the account $\mathrm{A1}$ is typically booked.

- Accounting rules are formalised in **propositional logics**, triggered by receipts properties.
- These rules can either be hard restrictions, or prone to exceptions.

Tensors: A natural description of factored systems

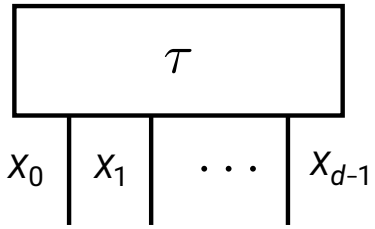
Factored System:[Russel and Norvig, 2021] Collection of variables $X_{[d]}$, each with a finite set of states $[m_k] = \{0, \ldots, m_k - 1\}$. The set of states is then the cartesian product

$$\bigtimes_{k \in [d]} [m_k]$$

Knowledge about the system: Store an information (e.g. Boolean) to each system state, and get Tensors

$$\tau\left[X_{[d]}\right] \; : \; \bigtimes_{k \in [d]} [m_k] \to \mathbb{R}$$

We represent them by diagrams with legs decorated by the variables:



Model Sparsity by Tensor Network Decompositions

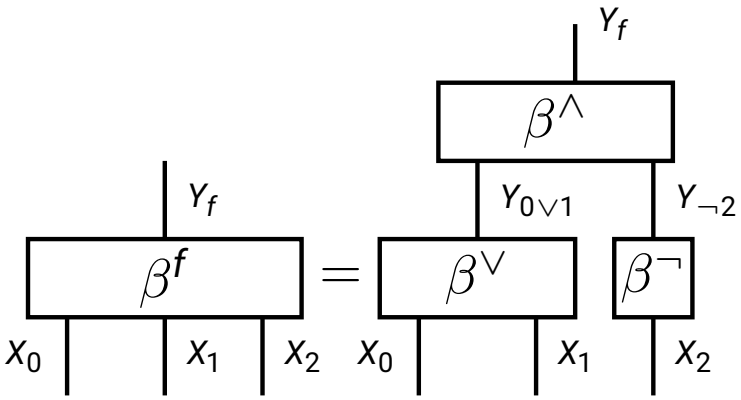
Basis encodings of functions between factored systems are tensors

$$\beta^q\left[Y_{[p]}, X_{[d]}\right] = \sum_{x_{[d]} \in \bigtimes_{k \in [d]} [m_k]} \epsilon_{q(x_{[d]})}\left[Y_{[p]}\right] \otimes \epsilon_{x_{[d]}}\left[X_{[d]}\right]$$

Main Sparsity Principle

Any decomposition of the function is reflected in a tensor network decomposition.

For the formula $f\left[X_{[3]}\right] = (X_0 \vee X_1) \wedge \neg X_2$, we have the following syntactical decomposition of its basis encoding:



Inference by Contractions

Reasoning: Amounts to contractions of the representations

- Probabilistic Inference: Contract graphical models for marginals
- Logical Inference: Deciding entailment based on model counts

Main Reasoning Principle

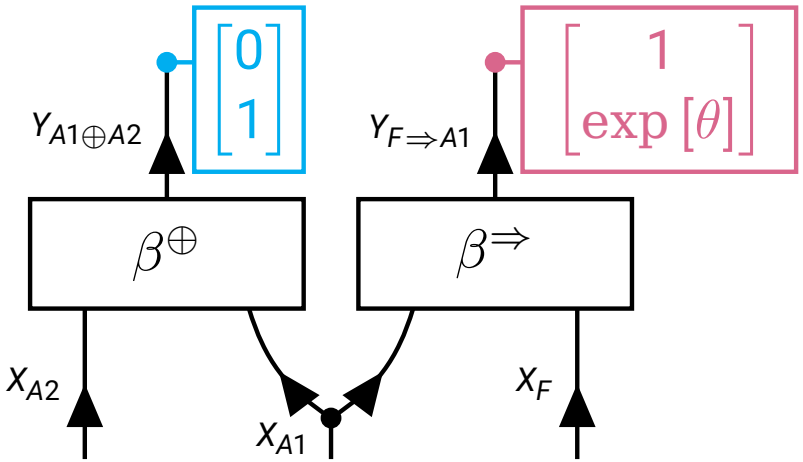
Model counts and marginal probabilities are contractions of tensor networks representing a model.

More advanced inference schemes:

- Parameter estimation: Moments in an inner loops, which are iteratively matched.
- Expectation propagation: Approximation framework for contractions by orchestration of efficient inference steps.

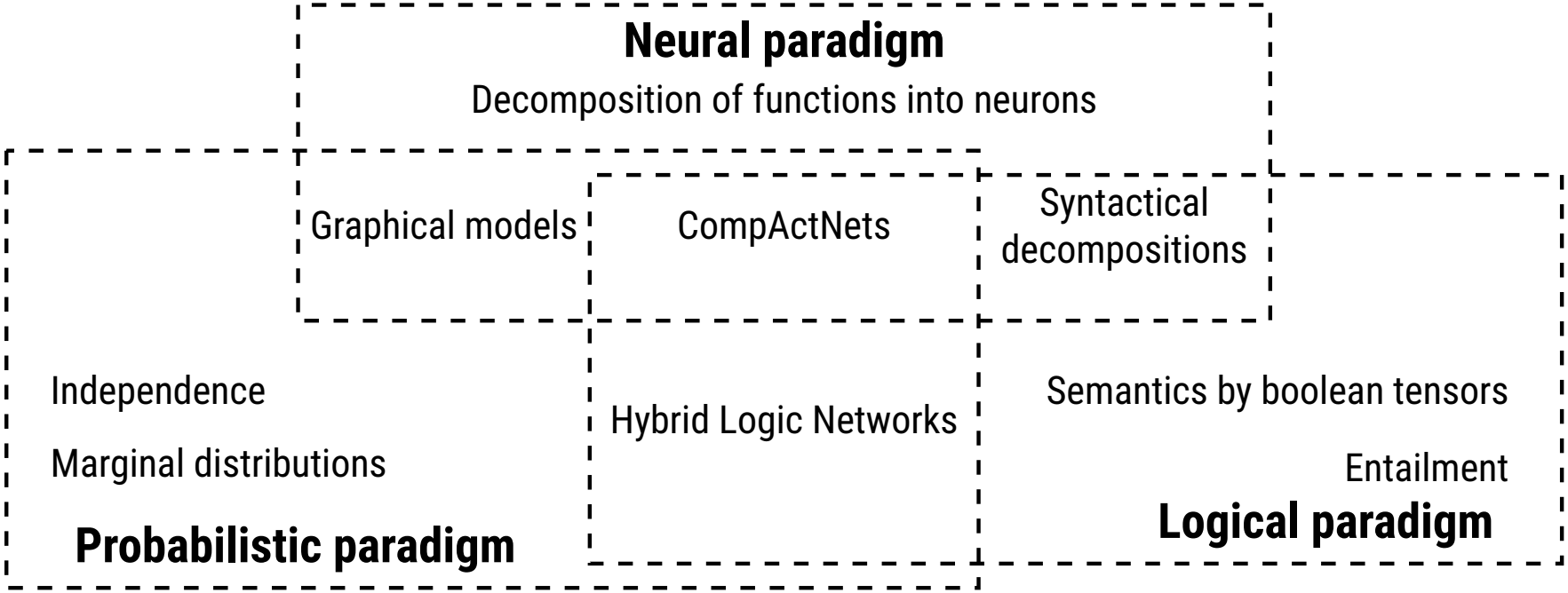
Toy Accounting Problem in a Tensor Network Representation

We represent the toy accounting model as a Computation-Activation Network with a trainable parameter $\theta \in \mathbb{R}$ enforcing the probabilistic second rule:



The contracted Computation-Activation Network is a tensor of order 3 in $\mathbb{R}^{2x2x2}$, with legs decorated by the variables X_{A1}, X_{A2}, X_F. Its coordinates are:

Tensor Networks for Neuro-Symbolic AI



Outline

- Tensor networks in neuro-symbolic AI
- Quantum Inference for Neuro-Symbolic AI

Motivation for Quantum Inference

Mitigate computational cost of contractions:

Get square root speedups through Quantum Inference.

The similarity of Quantum Circuits and Tensor Networks allows to apply quantum algorithms for inference tasks.

Quantum Rejection Sampling

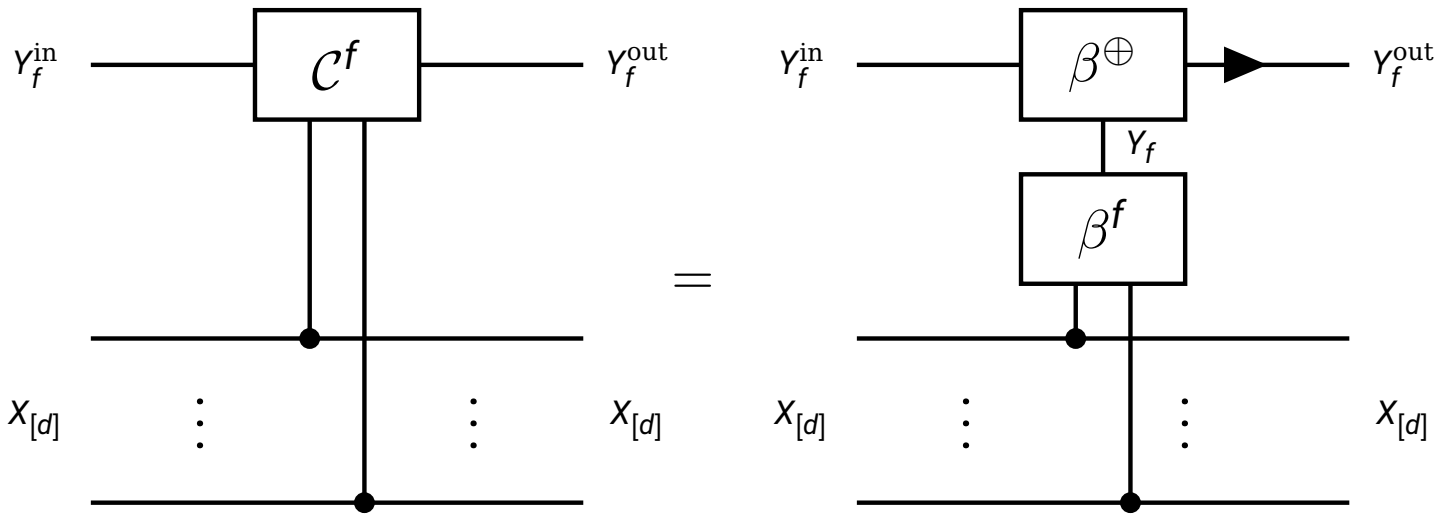
Square root advantage over classical rejection sampling schemes.

Tensor Networks to Quantum Circuits

Oracles, namely by $X_{[d]}$ controlled unitaries acting as

$$\left\langle \mathcal{C}^f \left[Y_f^{\text{in}}, Y_f^{\text{out}}, X_{[d]} = x_{[d]}\right], \epsilon_0 \left[Y_f^{\text{in}}\right] \right\rangle_{\left[Y_f^{\text{out}}\right]} = \epsilon_{f(x_{[d]})} \left[Y_f^{\text{out}}\right]$$

relate with basis encodings of functions as:



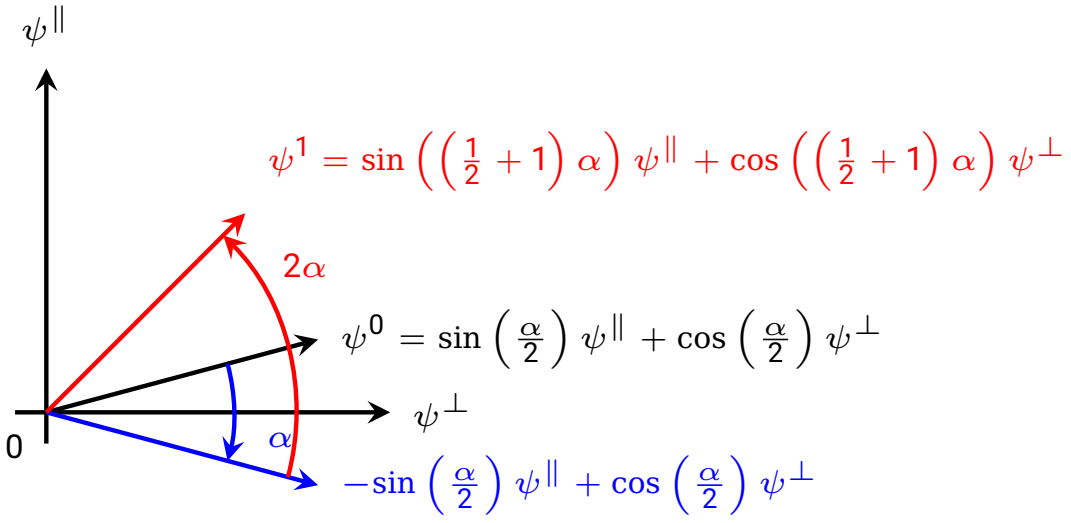
Same sparsity principles can be applied to get efficient quantum circuits for the oracles, and thus for the Grover operator.

Amplitude Amplification

Approach: Turn Tensor Networks into (conditioned) Bayes Network, introducing ancilla variables $A_{[p]}$

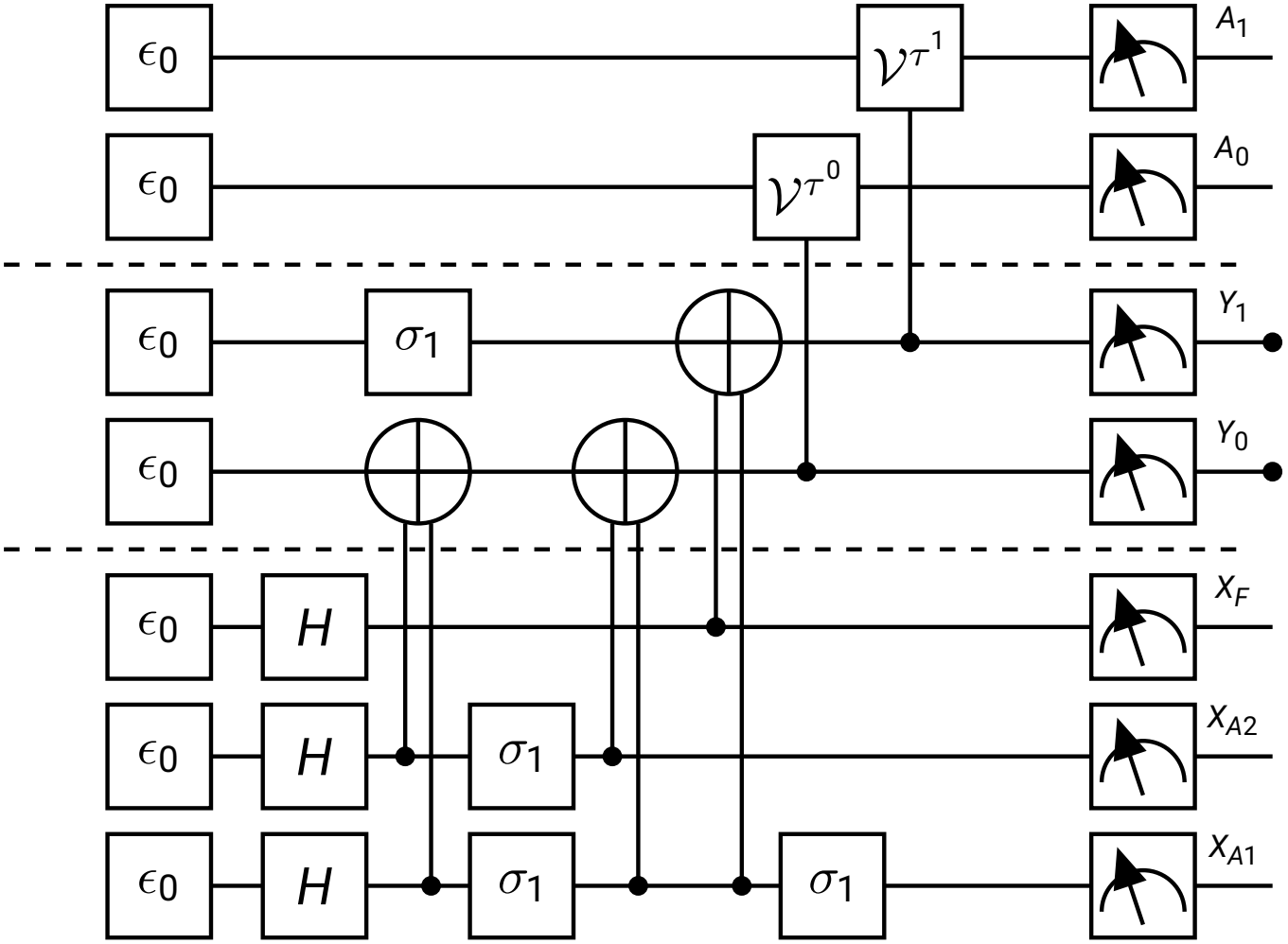
$$\lambda \cdot \mathbb{P}\left[X_{[d]}\right] = \tilde{\mathbb{P}}[X_{[d]}, A_{[p]} = 1_{[p]}]\,.$$

Amplify the sample acceptance event $A_{[p]} = 1_{[p]}$ to improve the classical acceptance rate $\lambda \in (0, 1]$, by application of the Grover operator.



Circuit for the Toy Accounting Problem

For the toy accounting problem, we prepare a distribution by the circuit:



Summary and Outlook

Tensor Networks appear as a natural framework for AI:

- Decomposition as Sparsity Principles
- Contractions as Inference

Quantum Circuits

- Follow the same tensor network formalism
- Allow for square root speedups for inference tasks